

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've been designing and building backend systems for 10 years, with the last 3 years focused on system architecture and cross-team technical leadership. I own API contracts that multiple teams depend on, design multi-account AWS architectures, and provide architectural guidance for a 12-person engineering team. I'm strongest in distributed systems, API design, and cloud infrastructure. I built a self-service platform that cut environment provisioning from weeks to hours and architected a system that won a 2024 Guidewire Innovation Award. I'm looking for Senior and Staff-level opportunities where I can own system-level problems that span teams.

Work Experience

Alberta Motor Association

Edmonton, Alberta

SOFTWARE ARCHITECT

Jan 2023 - Present

- Architected technical infrastructure for Canada's first pay-as-you-go auto insurance product across multiple AWS accounts with external partner integration. Product won 2024 Guidewire Innovation Award (1 of 3 worldwide), generated \$4.3M in annual revenue across 2.8M insured trips with 99.9% uptime. (AWS, Terraform, Lambda, Python)
- Maintain API architecture and design standards for insurance platform serving 5 internal teams and external partners. Approve new endpoint additions to ensure consistency. (Python, TypeScript, OpenAPI)
- Built self-service infrastructure platform that reduced environment provisioning from 2-3 weeks to 2 hours. Platform adopted by 12 engineers across 2 teams, averaging 10+ environments/month, and unblocked 5 major initiatives in 2025. Built solo. (Python, Terraform, AWS)
- Architected backend services for automated phone payment system, coordinating across 2 internal and 2 external teams. System improves advisor efficiency and customer experience. (Python, AWS, Stripe)
- Architected modernization of document generation system from legacy C# to containerized microservices on EKS. Led 2-3 engineers. System significantly improved reliability, observability, and operability, enabling environment creation in minutes (vs 2-3 weeks previously). (C# 8, Kubernetes, Datadog, Helm)
- Set coding standards and maintain technical documentation for 12-person team. Author RFCs for cross-team initiatives. Sit on quarterly Architecture Steering Committee that reviews and approves architectural proposals org-wide. (Merge remote-tracking branch 'origin/website')
- Currently driving adoption of trunk-based development with feature flags across team. Authored RFC proposing phased implementation to eliminate work stagnation, resolve environment bottlenecks, and achieve Guidewire compliance. Currently in pilot phase.

Arterys Inc. (Acquired by Tempus Labs)

Calgary, Alberta (Remote)

SOFTWARE DEVELOPER

Aug 2021 - Aug 2022

- Built backend services that ran medical AI algorithms and returned results to the platform. (TypeScript, Node.js)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching issues before customer impact.

Alberta Motor Association

Edmonton, Alberta

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Built automated test framework that reduced defects by 30% on subsequent projects. Framework now used org-wide across all Guidewire projects, still in use 4+ years later.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local dev environments.
- Led 5-6 person development team. Trained 5-6 developers on TDD methodologies.

Pleasant Solutions

Edmonton, Alberta

SOFTWARE DEVELOPER

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD. (C#)

Lockheed Martin Canada

Halifax, Nova Scotia

SOFTWARE ENGINEER ASSOCIATE

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging capabilities.

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.

Education

The Open University

MSC. COMPUTING (SOFTWARE ENGINEERING)

Currently pursuing Master's in Computing at The Open University.

Milton Keynes, England (Remote)

May 2024 - Ongoing

The King's University

B.S. IN COMPUTER SCIENCE

Edmonton, Alberta

Sep 2012 - Dec 2016

Skills

System Design

Design distributed systems with multi-account AWS architectures, Kubernetes, and Terraform. Built systems serving 5 teams that process \$4.3M in annual transactions with 99.9% uptime.

Technical Leadership

Own architectural decisions for 12-person engineering team. Approve API endpoint additions, write RFCs, and drive technical decisions that multiple teams follow.

Cross-Team Coordination

Coordinate across internal teams and external partners on multi-team initiatives. Designed systems used by 5 internal teams and external partners.

Platform Engineering

Built self-service infrastructure platform (60x faster provisioning) and test frameworks that reduced defects 30%. Focus on multiplying team effectiveness.

Technologies

Languages Python, TypeScript, Java, Bash

Backend REST APIs, OpenAPI, Microservices, CI/CD

Infrastructure AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM), Kubernetes, Docker, Terraform, Helm

Observability Datadog, Elasticsearch, Kibana

Databases PostgreSQL, Aurora/RDS, Redis